MILITARY SPECIFICATION SHEET

STARTER, AIRCRAFT ENGINE, AIR TURBINE, MODEL A-31

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the air turbine starter described herein shall consist of this specification and the latest issue of Specification MIL-S-19557.

APPLICABLE PARAGRAPH OF SPECIFICATION	
3.8.1	Envelope: See Figure 1 or 2 - Starter Figure 3 - Speed Switch
3.7	Weight, Max: Starter - 8.75 lb. (Wet) Speed Switch - 0.50 lb.
3.7.1	Overhung Moment, Max: 150 lb-in (Wet)
6.3.1	Aircraft Model: SH-60
4.5.16	Engine Model: T700
3.8.2.1	Quick-Attach-Detach Mounting: Four hole, 5.00 BC square flange in accordance with MS3331
3.8.2.2	Engine Accessory Drive: MS3326-2 (modified for speed)
3.8.3	Air Inlet and Exhaust Connections: (See Figure 1 or 2) The starter shall incorporate exhaust deflectors to direct starter exhaust in an aft direction. The deflectors shall be located on all exhaust openings from the 180° to the 300° positions as measured in a clockwise direction from top vertical center looking forward along starter horizontal centerline.

FSC 2995

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3.5.5
3.9.1 Rotation Viewed from Anti-Drive End:
Clockwise

3.3.2 Exposure Temperature Range (°F)
-65 to 160

3.3.2 External Surface Temperature: The external surface temperature of the starter shall not exceed 700°F during or following any operating condition. No external covering or insulating blanket(s) shall be used.

Control Valve:
3.3.2.3 Opening Rate, Max: 5 to 8 psig in the first 0.25 sec.
4.5.1.1 Opening Time, Max: 2.5 sec.
4.5.6 Closing Time, Max: 0.5 sec.
Navy Model: Not assigned

3.3.5 Lubrication: Addition or change of lubricant is permitted after 500 starter operating cycles or 500 hours of overrunning.

3.5.1 Rated Conditions: (Minimum allowable performance for 59°F, Sea Level Inlet Air Supply)

Output Drive Speed (rpm)	4500
Output Drive Torque (lb-ft):	16.2 Minimum
Airflow (lb per min):	26.3 Maximum
Air Inlet Total Pressure (psia)	45 Maximum
Air Inlet Total Temperature (°F):	400 Maximum

3.5.1
3.5.7 Cutout Speed: 15,000 to 17,000 rpm as controlled by the separate remote speed switch.

3.5.7 Automatic Shutoff Control: The speed switch shown on Figure 3 is a sealed unit and is separate from the starter. It is mounted away from the engine to provide protection from vibration and temperature extremes. A customer-furnished holding relay is required to complete the electric circuit. The speed switch functions to interrupt the starter control valve circuit when the starter reaches cutout speed. Input to the speed switch is provided by the engine start initiation switch and the starter monopole pickup. A single electronic speed switch has the capability of handling two engine/starter systems. The speed switch shall operate fully automatically for sequential or simultaneous engine starts and shall be completely compatible with all existing starters. The speed switch must be capable of switching a 1.0 amp inductive

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load at 20 to 32 volts DC for both starters. The starter monopole pickup signal, with an amplitude of 4 ±0.2 volts peak-to-peak at minimum frequency and a maximum amplitude of 20 volts peak-to-peak at all frequencies, shall be 26,300 to 28,000 Hz at starter cutout.

3.5.8 Altitude: The starter shall be capable of meeting rated speed and output torque with rated inlet air conditions up to and including 20,000 ft altitude. The starter shall not be damaged when subjected to 50,000 ft altitude in the non-operational/overrunning mode.

3.5.9 Attitude: The starter shall be capable of meeting the operational requirements of this specification when the starter centerline, taken through the output shaft, is any position within 10 degrees of the horizontal position, unless specified otherwise on this Specification Sheet.

4.2.2 Inspection/Acceptance Test: Additional test requirements are:

(1) The duration of the overrunning test shall be 10 minutes;

(2) Perform 4 engine start endurance cycles at rated inlet conditions;

(3) Determine the dielectric strength of the speed pick up;

(4) Check turbine wheel integrity at maximum operating speed;

(5) Check proof pressure of all pressurized components at maximum operating pressure;

(6) Check for oil leakage.

4.5.4 Initial Calibration: (at 59°F, Sea Level Inlet Air Conditions)

DRIVE TORQUE LB-FT	DRIVE SPEED RPM	DRIVE SPEED CONDITION	AIRFLOW LB/MIN
20.8 MIN	0	STALLED	26.3 MAX
16.2 MIN	4,500	RATED	26.3 MAX
5.6 MIN	15,000	CUTOUT	26.3 MAX

4.5.5 No-Load Operation

Time:	60 Seconds
Output Drive Speed:	19,560 rpm MINIMUM

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3.3.5
4.5.6 Endurance Test: Total operation of 2,000 cycles consisting of: 1,925 engine starts; 50 engine thermal stabilization runs, one run following each 40 engine starts; 25 motoring runs, one run following each 80 engine starts. Cycles to be conducted as follows:

(1) Engine Start. Each start to consist of three phases as shown below:

TEST STAND ROTOR POLAR MOMENT OF INERTIA	PHASE A Acceleration		PHASE B Stabilized Conditions			PHASE C Acceleration	
	Drive Speed RPM	Time SEC	Drive Speed RPM	Drive Torque LB-FT	Time	Drive Speed RPM	Time SEC
LB-FT^2	±50	(MAX)	±50	(MIN)	SEC	(MIN)	(MAX)
3.54	4,500	8	4,500	16.2	10	15,000 (cutout)	20

(2) Engine Thermal Stabilization Run: Operate starter to accelerate the 3.54 lb-ft^2 inertia load to 7,800 rpm starter output speed and apply a torque load to maintain this speed for 90 seconds. Remove torque load and allow starter to accelerate inertia load to cutout speed. Wait at least 30 minutes before next starter operation.

(3) Motoring Run:

(a) Motor starter (with 3.54 lb-ft^2 inertia load and torque load) at 7,800 rpm output speed for one minute and shutdown for 15 minutes;

(b) Motor for 2 minutes and shutdown for 5 minutes;

(c) Motor for 1 minute and shutdown for 15 minutes;

(d) Motor for 2 minutes and shutdown for 5 minutes;

(e) Motor for 30 seconds, then remove torque load to allow starter to accelerate the inertia load to cutout speed;

(f) Wait 30 minutes before next starter operation.

4.5.8 Rated Torque Calibration:

Minimum Output Drive Speed: 4,500 rpm
Maximum Airflow: 26.3 lb per min.

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3.3.5
4.5.9 Overrunning: The starter output drive shall be driven at 30,500 rpm for 3,000 hours in a clockwise direction (viewed from the anti-output shaft end) in accordance with the following tabulation. The ambient air temperature and attitudes shall vary in accordance with the following tabulation while the starter pad temperature is maintained at 225 ±10°F. The attitude transitions are to be completed within 30 seconds.

Time Minutes	Attitude ±2° (see Note)	Ambient Temperature +10°F
4	0°	130
6	+45°	130
4	0°	130
1	+90°	130
30	0°	275
1	-90°	130
4	0°	130
6	-45°	130
4	0°	130

Note: Position of the starter anti-drive end with respect to the horizontal centerline.

Lubricant addition or change is permitted after each 500 hours of overrunning.

4.5.10 Extreme Temperature Operation:

NO. OF CONSECUTIVE STARTS	AMBIENT TEMPERATURE °F ±5°	INLET AIR CONDITIONS	
		TOTAL PRESSURE PSIA ±0.5	TOTAL TEMPERATURE °F ±10°
4	-40	44.7	275
2	160	44.7	450

4.5.11 Consecutive Cyclical Operation:

Number of cycles: 4 below 60°F
2 from 60 to 125°F

4.5.12 Sustained Motoring:
Time (Minutes):
2 on
5 off
2 on
30 off

The starter shall be capable of motoring operation in accordance with the endurance cycles of paragraphs 4.5.6(2) and (3).

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4.5.13 Vibration: MIL-STD-810B, Method 514, Procedure I (Parts 1, 2, and 3), Figure 514-1, Curve M.

Burn-In: Each component, which includes an electronic part, shall be subject to the following burn-in procedure, which includes six complete temperature cycles between -67 to +158°F and ten minutes of vibration while stabilized at each temperature extreme. Complete functional testing shall be performed at each temperature extreme. Temperature, vibration, functional testing, and time requirements are as follows:

a. Temperature extremes shall be maintained at -71 to -63°F (-57 to -53°C) on cold cycles and 154° to 162°F (68 to 72°C) on hot cycles.

b. The time required on the cold cycle shall include twenty minutes minimum stabilization after reaching the cold temperature range and before beginning vibration and functional testing. Transition from hot to cold shall not exceed one hour.

c. The time required on the hot cycle shall include one hour and forty minutes minimum stabilization after reaching the hot temperature range and before beginning vibration and functional testing. Transition from cold to hot shall not exceed forty minutes.

d. Units shall be non-operative during transition and stabilization on the cold temperature cycles. They shall be operative during transition and stabilization on the hot cycles.

4.5.15.1 Slip Clutch Torque: 575 lb-in. (MAXIMUM)

4.5.15.2 Shear Section Strength: 500 to 575 lb-in.

4.5.18 Containment Tests: Applicable

4.5.18.1 Hub Containments Tests:

Inlet Air Pressure:	45 psia
Inlet Air Temperature:	375°F

4.5.18.2 Failure at No-Load:

Inlet Air Pressure:	66 psia
Inlet Air Temperature:	480°F

3.4 Reliability and Maintainability.

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3.4.1 Reliability Program: The manufacturer shall prepare a proposed reliability program plan in accordance with MIL-STD-785 and shall submit it to the Qualifying Activity. Qualification approval will not be granted to a product until a reliability program for the product has been approved by the Qualifying Activity, in addition to the product's passing the Qualification tests. The reliability design requirements shall be as follows:

Corrective Maintenance Frequency. The mean-time-between-failures for the starter shall be not less than either 6,000 start cycles or 6,000 engine/starter overrunning hours during the useful life.

Removal Frequency. The mean-time-between-on-aircraft removal due to a corrective maintenance action shall be not less than either 6,000 start cycles or 6,000 engine/starter overrunning hours during the useful life.

Useful Life. The starter shall have a minimum useful life of 8,000 operating hours.

Storage. The pneumatic starter shall have a minimum total shelf life of five (5) years when stored as specified by the contractor under original packaging conditions.

3.4.2 Maintainability Program. The manufacturer shall prepare a proposed maintainability program plan in accordance with MIL-STD-470 and shall submit it to the Qualifying Activity. Qualification approval will not be granted to a product until a maintainability program for the product has been approved by the Qualifying Activity, in addition to the product's passing the Qualification tests. Maintainability design requirements shall be as follows:

Corrective Maintenance

a. The item shall be capable of being removed and replaced by not more than 1 man in not more than 20 elapsed minutes excluding item access time.

b. Depot level repair shall not require more than 16 elapsed hours using not more than one man.

Scheduled Removals. There shall be no scheduled removals required.

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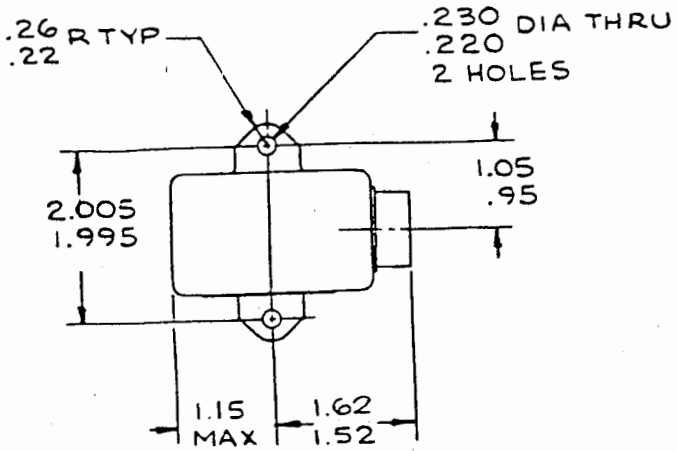
3.3.8 Interchangeability: The ATSCV shall be physically interchangeable or replaceable with any other manufacturer listed on the applicable Qualified Products List (QPL).

PREPARING ACTIVITY
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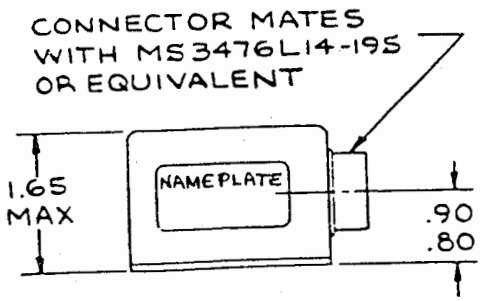
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PIN	FUNCTION
A	MONO. SHLD. #1
B	ELECT. SYST. GRD.
C	+ DC POWER #1
D	SILR (RELAY) #1
E	MONO. #1
F	MONO. #1
G	MONO. SHLD. #2
H	ELECT. SYST. GRD.
J	+ DC POWER #2
K	SILR (RELAY) #2
L	MONO #2
M	MONO. #2
N, P, R, S, T, U, V	NOT USED



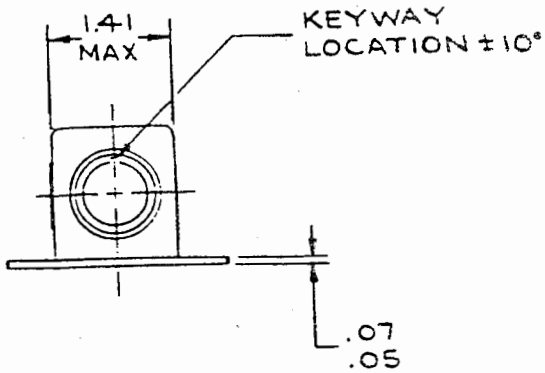


FIGURE 3. Starter speed switch.

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